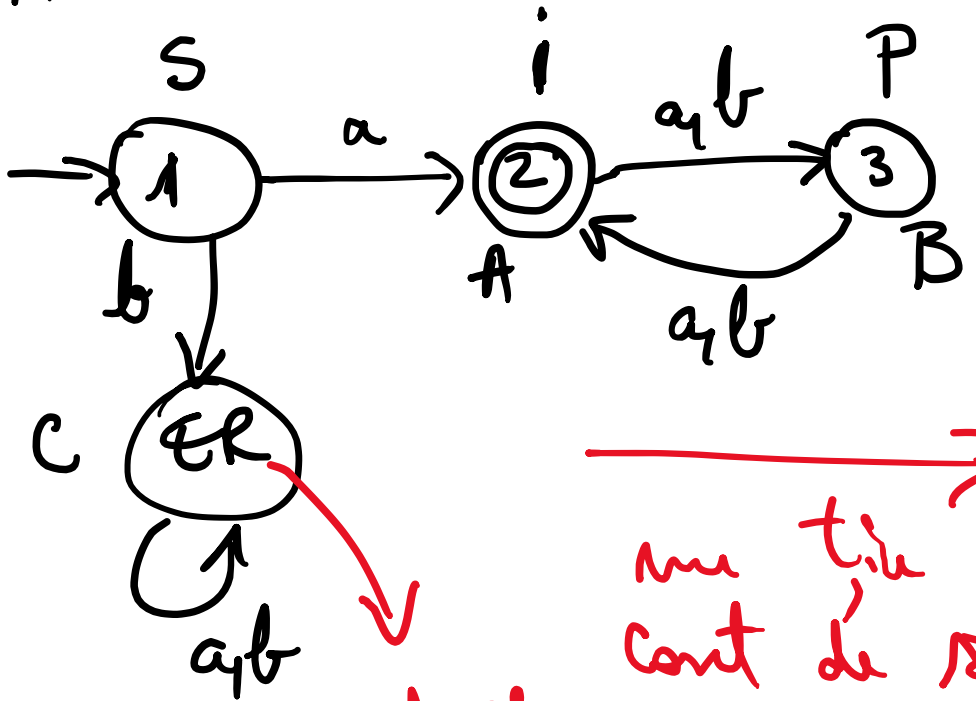
$\{w \in \{a,b\}^* \mid |w| = \text{impară}\}$ w începe cu a $\gamma = LR$

AFD $\Rightarrow$ GR



$S \to aA \mid bC$

$A \to aB \mid bB \mid e$

$B \to aA \mid bA$

$C \to aC \mid bC$

nu poate genera
nicio stare terminală

GR $\Rightarrow$ ER ?

GR => ER

$S \to aA \mid \cancel{bC}$

$A \to aB \mid bB \mid e$

$B \to aA \mid bA$

~~$C \to aC \mid bC$~~

$S \to aA$

$A \to aB \mid bB \mid e$

$B \to aA \mid bA$

$S : x$

$A : y$

$B : z$

$\Rightarrow$

$$\begin{cases} x = ay \\ y = az + bz + e \\ z = ay + by \end{cases} \Rightarrow x = ?$$

$z = (a+b)y \Rightarrow y = (a+b)(a+b)y + e$

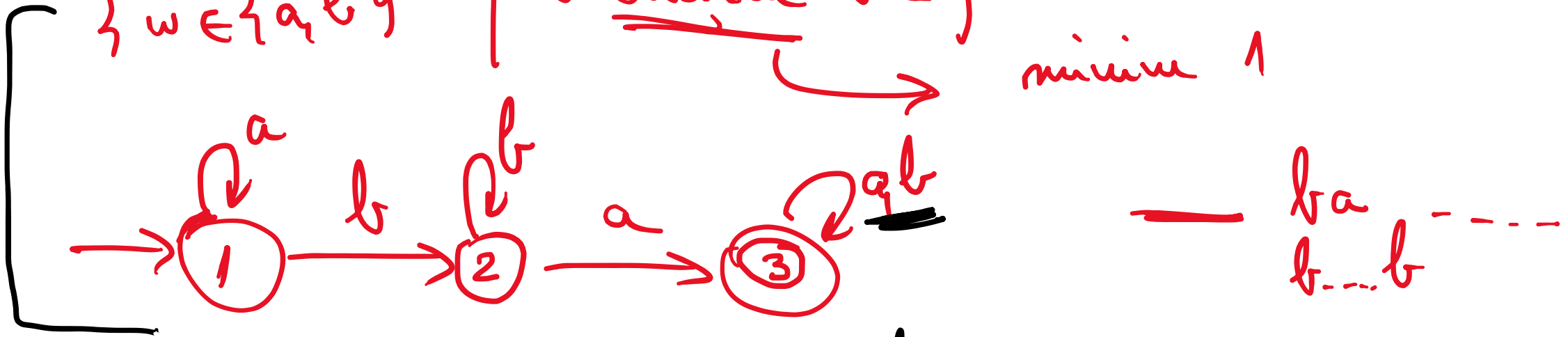
$y = ((a+b)(a+b))^* \Rightarrow x = a((a+b)(a+b))^*$

2) $\{w \in \{a,b\}^* \mid w$ conține un număr impar de $ba\}$

AFD

$\{w \in \{a,b\}^* \mid w$ conține $ba\}$

minim 1



— ba

$b \ldots b$ - - - -

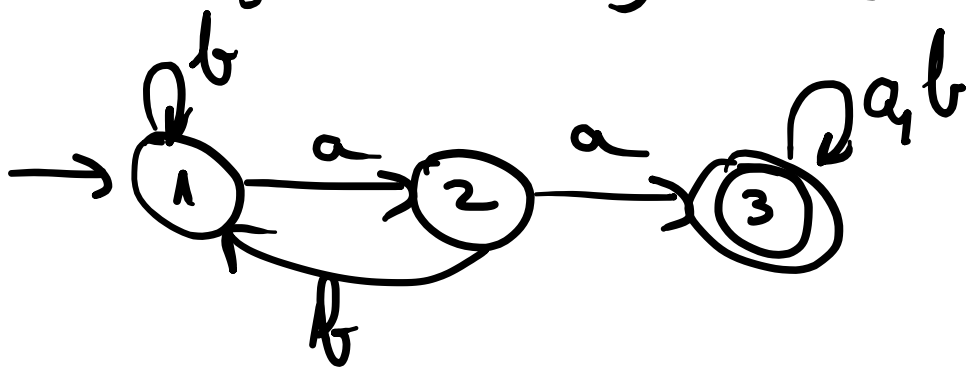
w conține un număr impar de ba

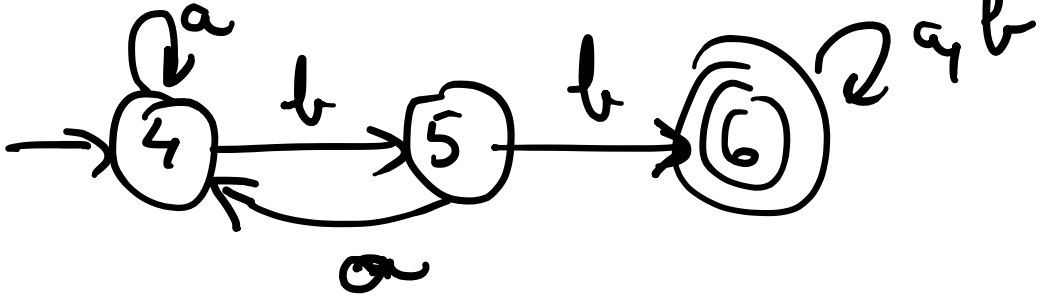


... ba

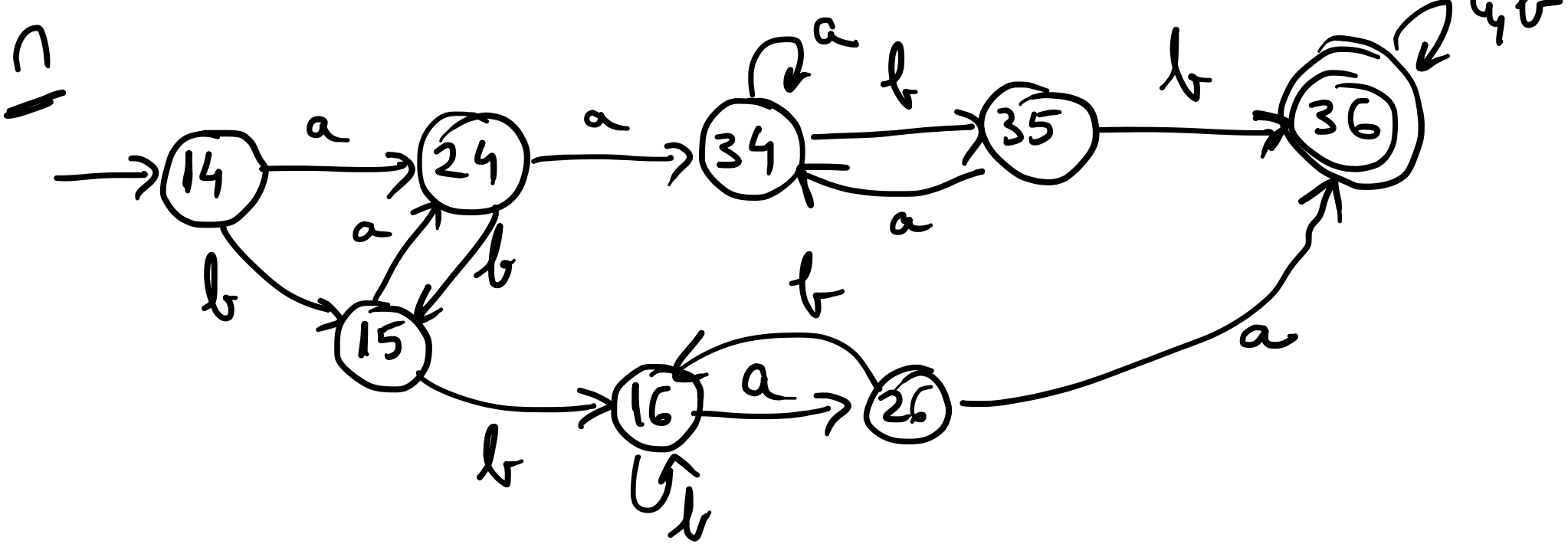
$\{w \in \{a,b\}^* \mid w$ conține aa și bb$\}$ ATD

$\{w$ conține aa$\} \cap \{w$ conține bb$\}$

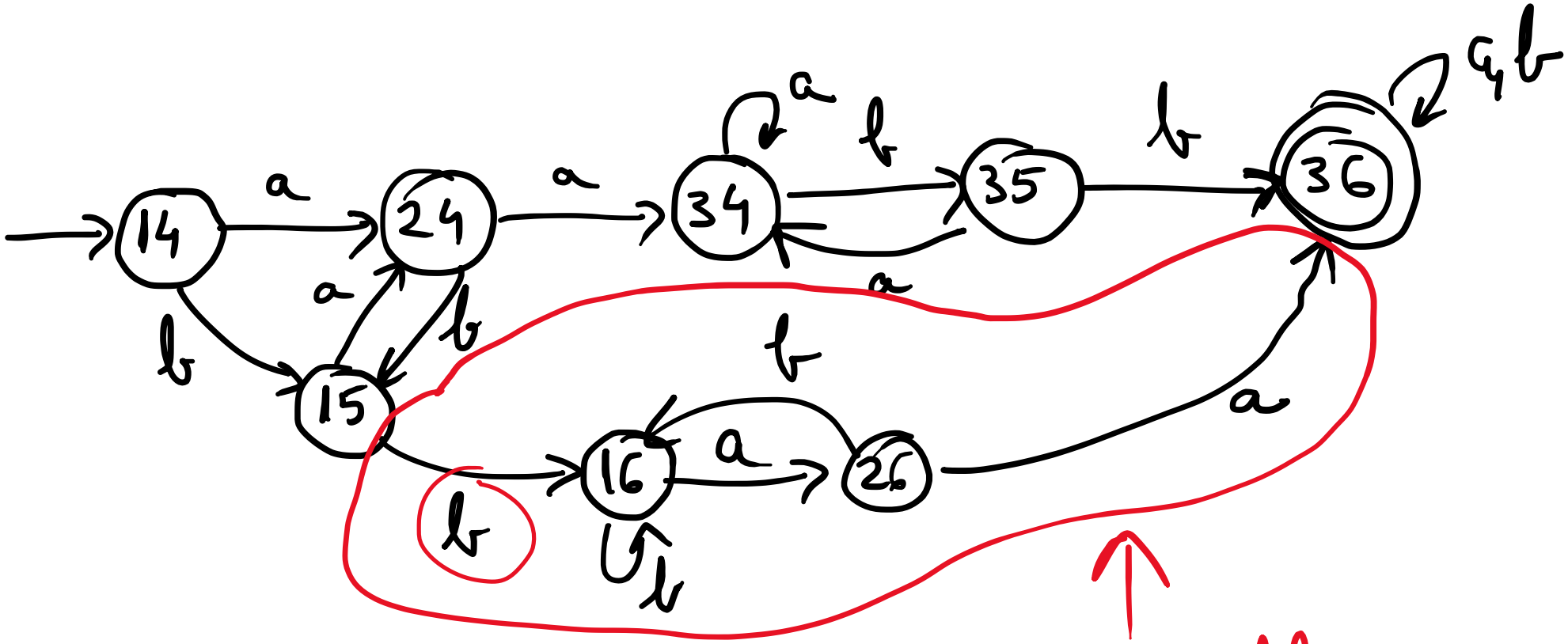




$\cap$

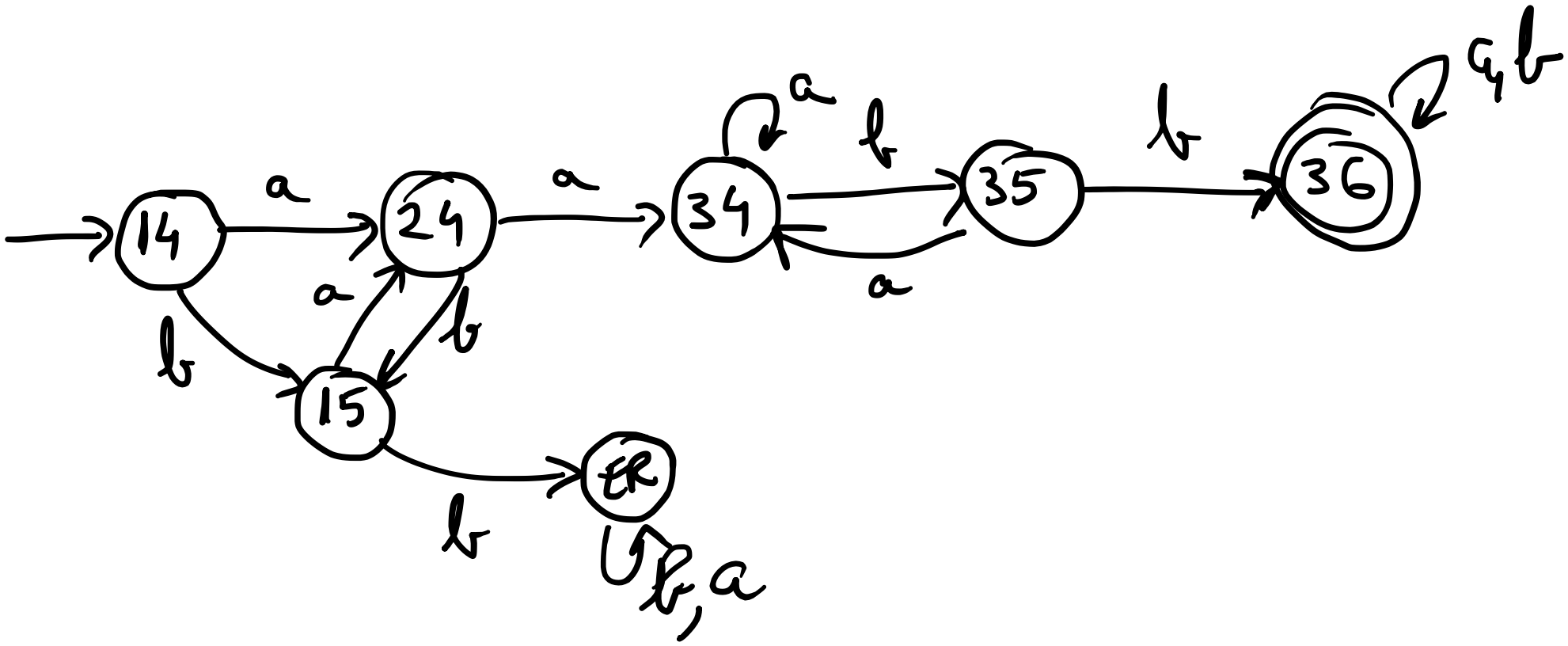


$\{w \in \{ab\}^* \mid w$ conține aa și bb, prima apariție a aa înainte de $bb\}$



AFD w conține aa și bb

$\{w \in \{a,b\}^* \mid w$ conține aa și bb, prima apariție aa înainte de $bb\}$

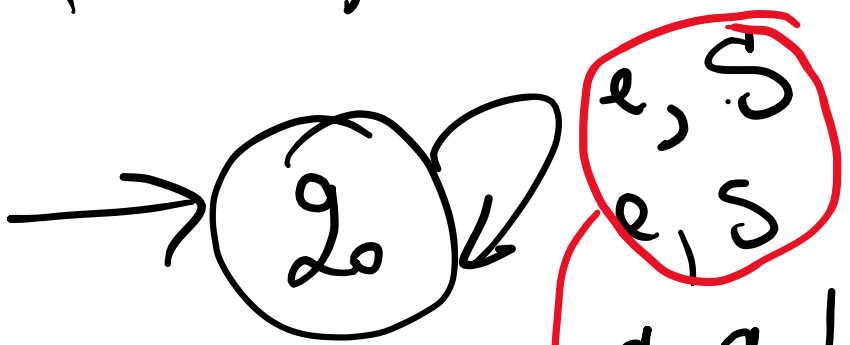


$\{ b^{2n} a^{3n} \mid n \geq 1 \}$ → construiți acceptorul

= LIC ⇒ acceptorul = AS

$(bb)^n (aaa)^n$ $\quad S \to bbSaaa \mid bbaaa$

Producții ⇒ AS = acceptare prin stivă goală



$bbSaaa \quad \not\in \quad S \to bbSaaa$

$bbaaa \quad \not\in \quad S \to bbaaa$

$* \; x \in \{a,b\}^*$

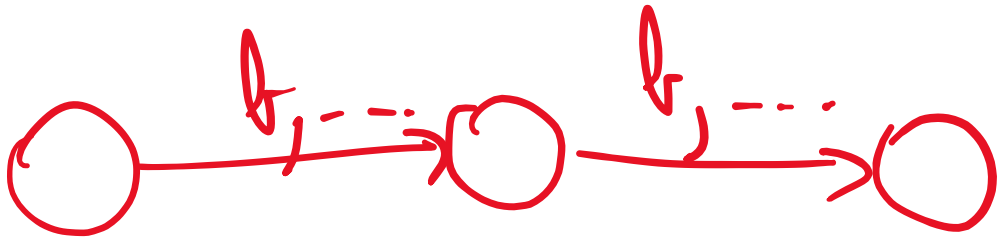
⇓ ASD / $\mathcal{H}$?

ASH

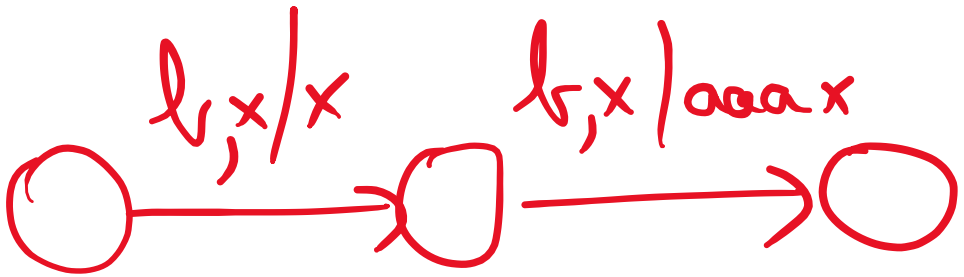
$b^{2n} a^{3n}$ $n \geq 1$ AS fără gram

2-b $\Rightarrow$ pun în stivă 3-a





2-b consecutive

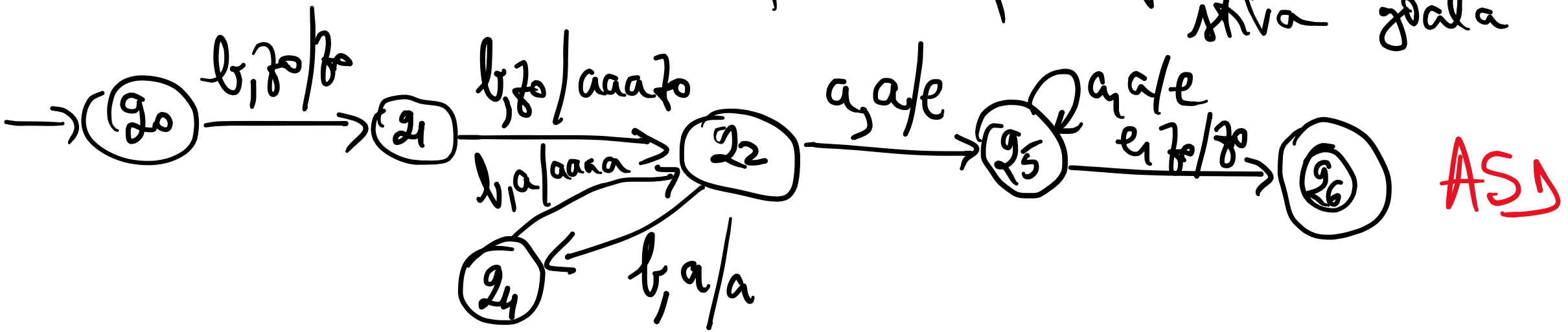


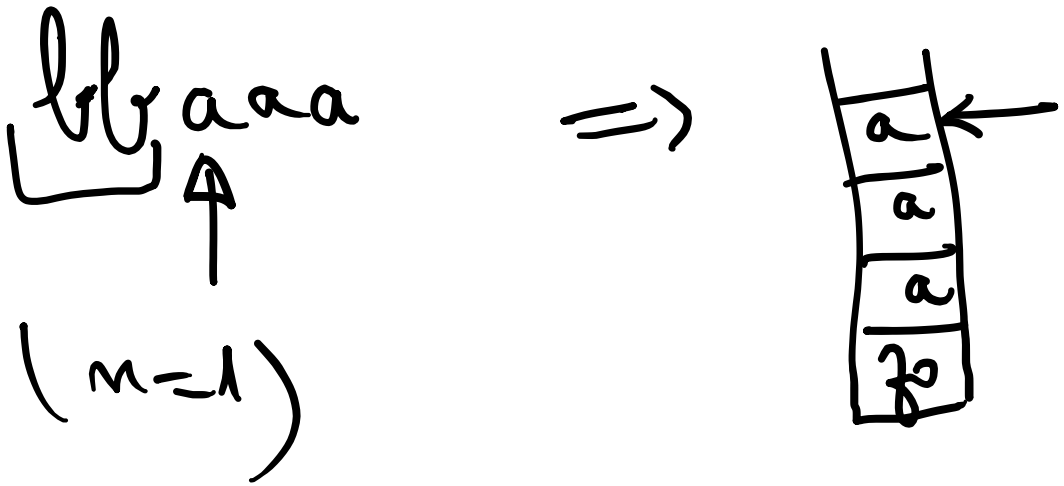
2-b consecutive

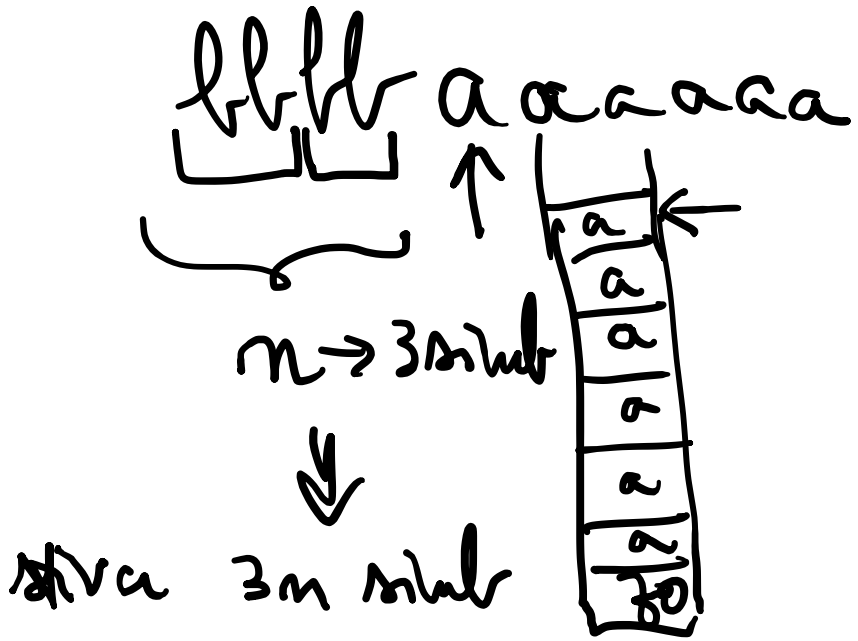
$L = \{ b^{2n} a^{3n} \mid n \geq 1 \}$

AS → începe cu în stivă Z_0

AS nu poate funcționa dacă stiva goală

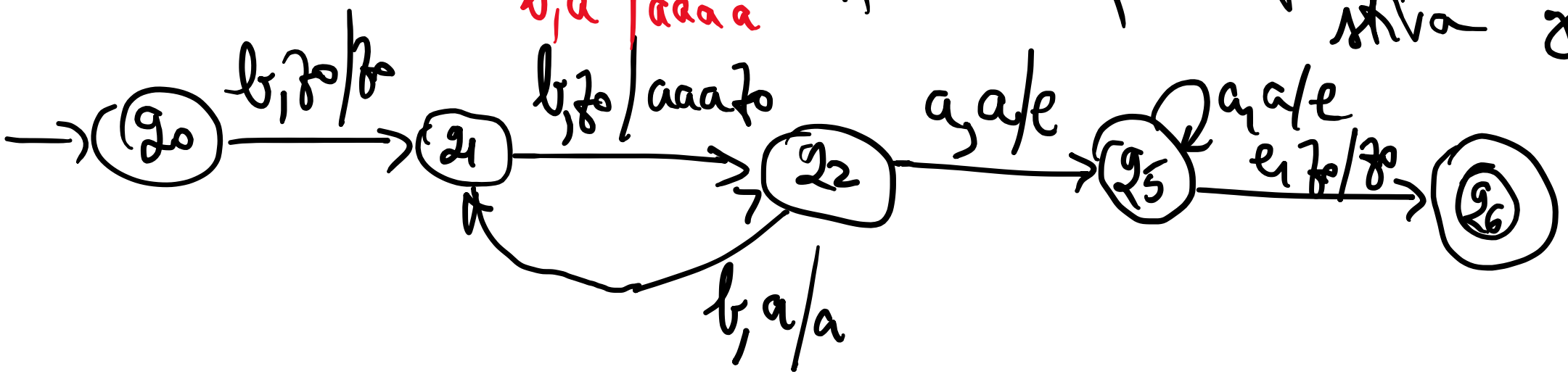






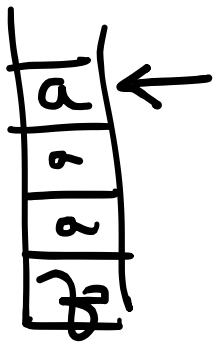
$L = \{a^{2n} b^{3n} \mid n \geq 1\}$

AS → începe cu în stivă Z_0
AS nu poate funcționa dacă stiva goală



ASD

bb bbbaa - - - -



ASD: 1) $|m(q, x, y)| \leq 1$

2) dacă $|m(q, x, y)| = 1$

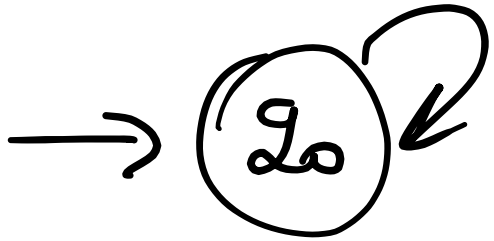
$\Downarrow$

$m(q, x, y) = \phi \quad \forall x \in \Sigma$

$\{ a^{2n} (bc)^n \mid n \geq 1 \}$

$S \rightarrow aaSbc \mid aabc$

AS



ASA

$\epsilon, S \mid aaSbc$

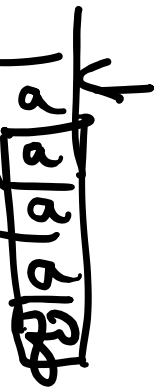
$\epsilon, S \mid aabc$

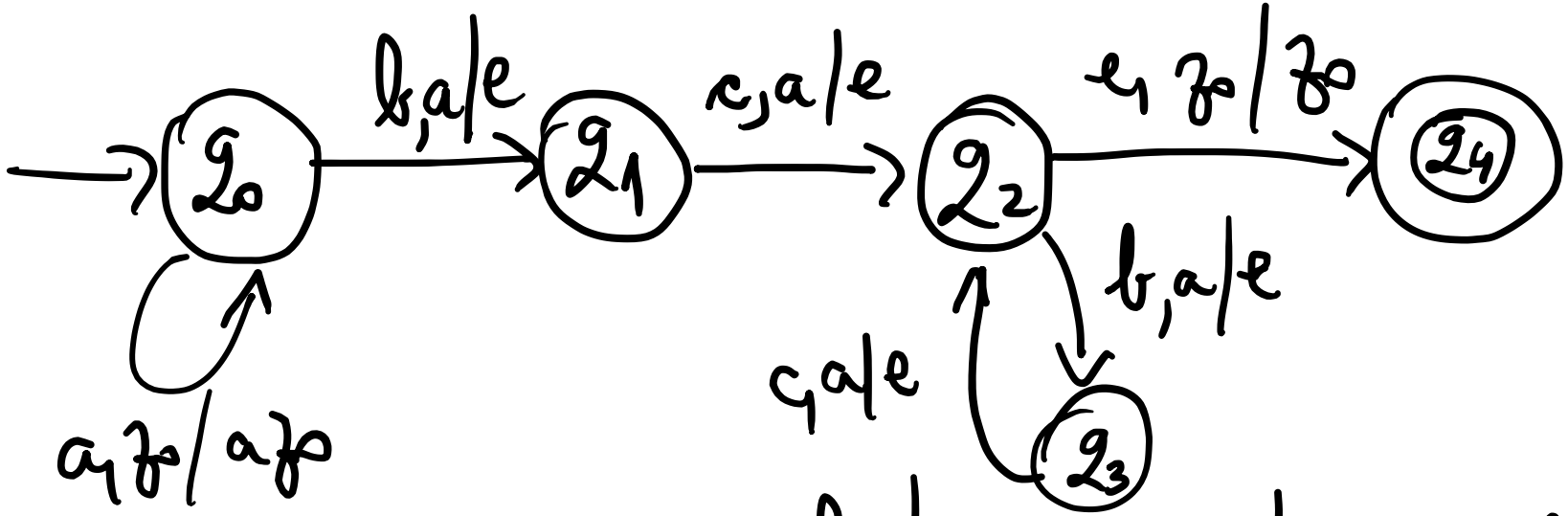
$x, x \mid \epsilon$

$\forall x \in \{a, b, c\}$

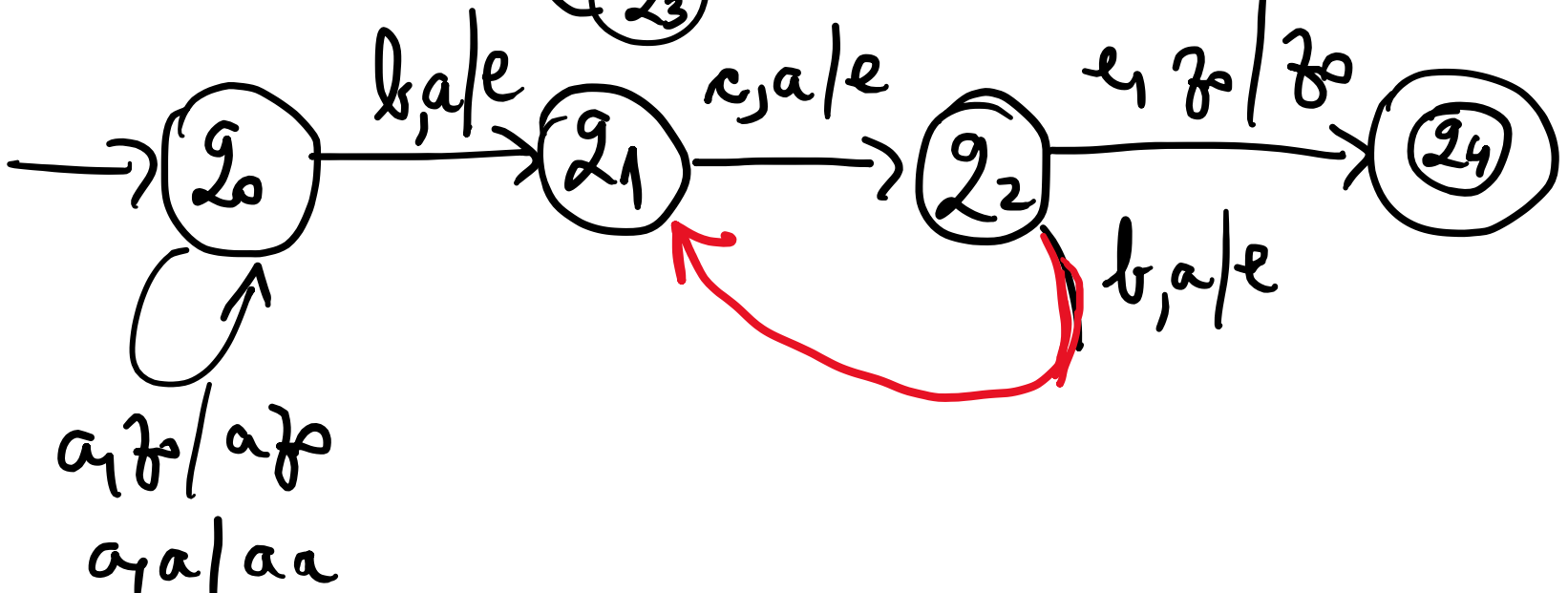
$a^{2n}(bc)^n$ $n \geq 1$

aaaa bcbc





ASD

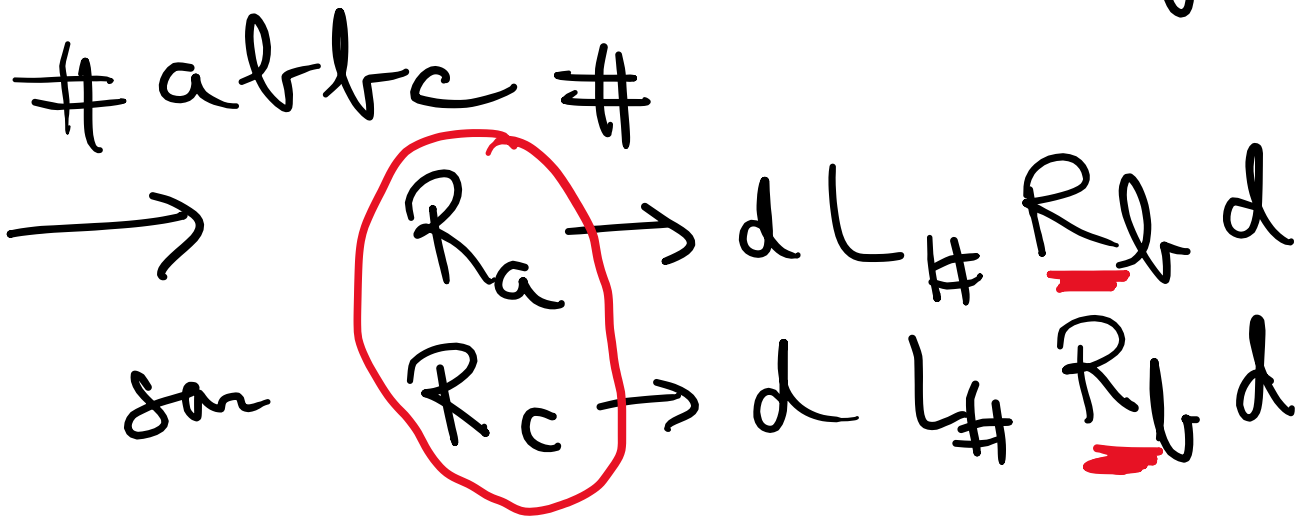


pour

$\{w \in \{a,b\}^* \mid \#_a(w) + \#_c(w) = \#_b(w)\}$

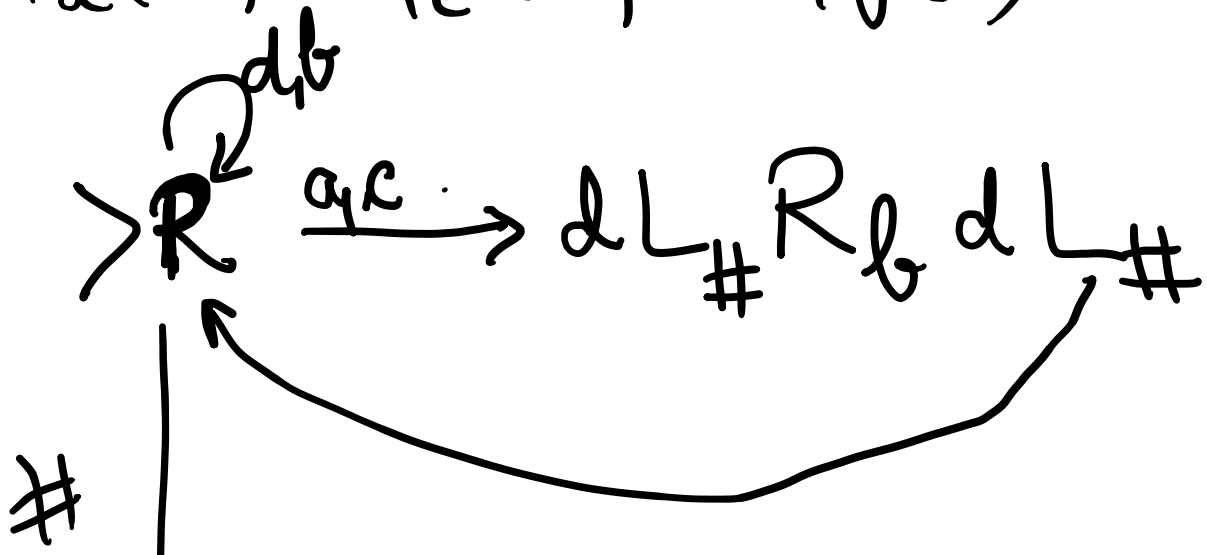
MT accepta limbajul

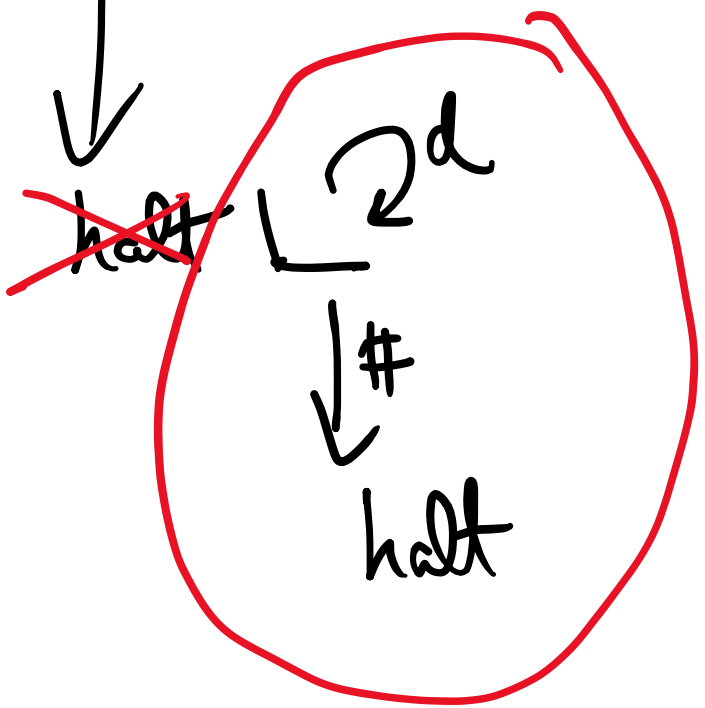
#abbc#



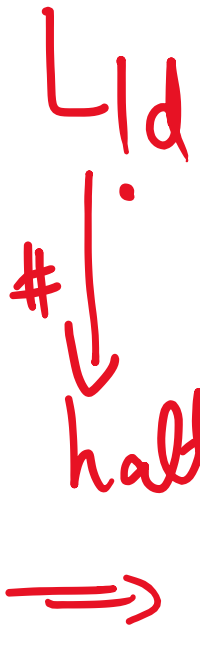
se repetă până am marcat tot

$\#_a(w) + \#_c(w) = \#_b(w)$



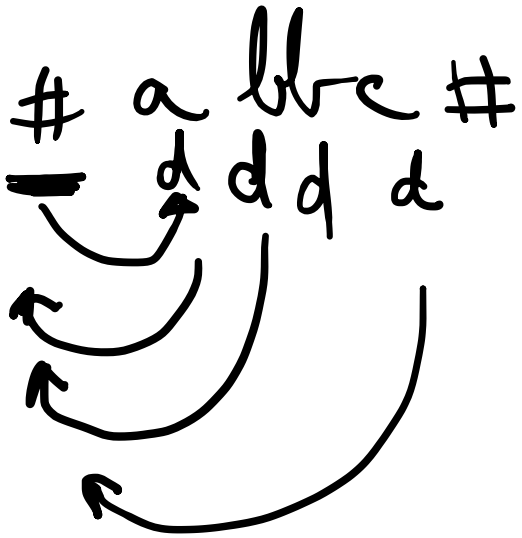


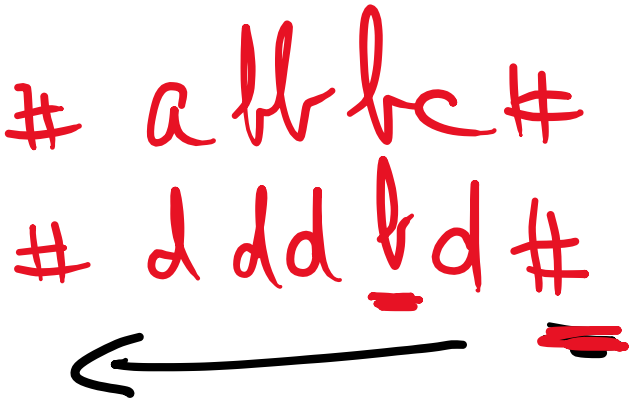
Sau



#w# $\Longrightarrow$ final

#w#





#w#